

Smart City Civic Complaint and Management System

Software Engineering Project Journal – 02

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Journal 02

Gantt Chart and Use Case Diagram

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1. Objective

The objective of this journal is to document the project planning and functional modelling activities completed after the project proposal.

The main objectives were:

- To convert the proposed project timeline into a Gantt Chart aligned with the course W1–W20 schedule.
- To identify the major actors who will interact with the system.
- To identify the major use cases and responsibilities associated with each actor.
- To represent the system boundary and actor–system interactions using a Use Case Diagram.
- To organize the completed artifacts in the project repository for future development and documentation.

2. Work Completed

After completing the Project Proposal documented in Journal 01, the next project activity was system planning and functional modelling.

Two major Software Engineering artifacts were completed:

1. Gantt Chart
2. Use Case Diagram

These artifacts were created before starting implementation so that the development work has a defined schedule and a clear understanding of the system's functional interactions.

3. Gantt Chart

3.1 Purpose

A Gantt Chart was prepared to convert the project plan into a week-by-week development schedule.

The chart is aligned with the course timeline from W1 to W20. It represents activities such as project planning, requirements analysis, UML and system design, core development, prototype testing, improvement, advanced features, final testing, and final submission.

3.2 Planning Approach

The project follows the incremental approach defined in the proposal. The core complaint workflow is prioritized first, while advanced smart features are planned for later increments.

The major academic milestones represented in the chart are:

- W4 – Project Proposal + Report Submission
- W7 – Prototype + Report Submission
- W8 – Improvement Plan
- W9–W10 – MST Week
- W11–W12 – Progress Monitoring
- W13 – Buffer
- W14 – Progress Monitoring
- W15 – Diwali Break
- W16 – Progress Monitoring over the second prototype
- W17 – Final Prototype + Presentation + Report Submission
- W18 – Assessment Week (Buffer)
- W19–W20 – EST Week

The Gantt Chart also allows activities to overlap when they can be performed in parallel. For example, design activities and preparation for development can occur during the same period.

3.3 Outcome

The Gantt Chart provides a clear schedule for the project and helps identify when each major activity is expected to be performed. It also provides a reference for monitoring progress in later journals.

4. Use Case Diagram

4.1 Purpose

A Use Case Diagram was prepared to model the functional relationship between the system and its external actors.

The proposed system connects citizens, department officers, field workers, and administrators through a structured complaint-management workflow.

4.2 Identified Actors

Actor	Main Responsibilities
Citizen	Register/login, submit complaints, track complaint status, verify resolution, provide feedback, and reopen complaints.
Department Officer	Review and verify complaints, assign workers, monitor progress, and manage complaint status.
Field Worker	View assigned tasks, accept work, update work progress, upload evidence, and mark work as completed.
Administrator	Manage users, departments, workers, and monitor complaints.

4.3 Main Use Cases

The major use cases represented in the diagram include:

- Register / Login
- Submit Complaint
- Track Complaint Status
- Verify Resolution
- Provide Feedback
- Reopen Complaint
- Select Category
- Enter Description
- Provide Location
- Generate Complaint ID
- Upload Photograph
- Review Complaint
- Verify Complaint
- Assign Worker
- Monitor Progress
- Manage Complaint Status
- View Assigned Tasks

- Accept Work
- Update Work Progress
- Upload Evidence
- Mark Work Completed
- Manage Users
- Manage Departments
- Manage Workers
- Monitor Complaints

The diagram also represents the planned Smart Assistance layer containing Smart Classification, Duplicate Detection, and Priority Calculation. These features are planned for later increments after the core workflow is stable.

5. Core Workflow Identified

The functional model establishes the following main workflow:

Citizen → Complaint Submission → Officer Verification → Worker Assignment → Work and Evidence → I

This workflow is consistent with the project proposal's objective of creating a complete and traceable path from complaint creation to resolution and citizen verification.

6. Repository Organization

The completed artifacts were organized in the repository using separate folders for visual assets and documentation:

```
assets/
  gantt-chart.png
  use-case.png
```

```
docs/
  gantt-chart.pdf
  use-case.pdf
```

The PNG files are used as visual assets, while the PDF versions provide printable/documentation copies.

7. Result

The following work has been completed in this journal:

Activity	Status
Project timeline planned	Completed
Gantt Chart prepared	Completed
System actors identified	Completed
Major use cases identified	Completed
Use Case Diagram prepared	Completed
Artifacts organized in repository	Completed

8. Learning / Engineering Understanding

This activity helped establish that Software Engineering involves planning and modelling the system before implementation.

The Gantt Chart provides a project-management view of *when* work will be performed, while the Use Case Diagram provides a functional view of *who* interacts with the system and *what* they can do.

These artifacts will be used as inputs for the next stages of requirements and system design.

9. Conclusion

Journal 02 documents the completion of the Gantt Chart and Use Case Diagram for the Smart City Civic Complaint and Management System.

The project now has a defined W1–W20 development schedule and a functional model covering citizens, department officers, field workers, and administrators. The core complaint workflow has been identified, while advanced smart features remain planned for later increments.

The next project activities will build on these models rather than changing the already established project scope without justification.